

Using Python Programming

1. Hospital Patient Admission Analysis

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    'Department': ['Cardiology', 'Neurology', 'Pediatrics', 'Orthopedics', 'Emergency'],
    'Admissions': [320, 250, 410, 280, 520]
}

df = pd.DataFrame(data)

plt.figure(figsize=(8,5))
sns.barplot(x='Department', y='Admissions', data=df)
plt.title('Patient Admissions by Department')
plt.show()

monthly = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr'],
    'Cardiology': [70, 80, 85, 85],
    'Neurology': [55, 60, 65, 70],
    'Pediatrics': [90, 100, 110, 110],
    'Orthopedics': [60, 70, 75, 75],
    'Emergency': [120, 130, 135, 135]
})

monthly.set_index('Month').plot(kind='bar', stacked=True)
plt.title('Monthly Department Admissions')
plt.show()

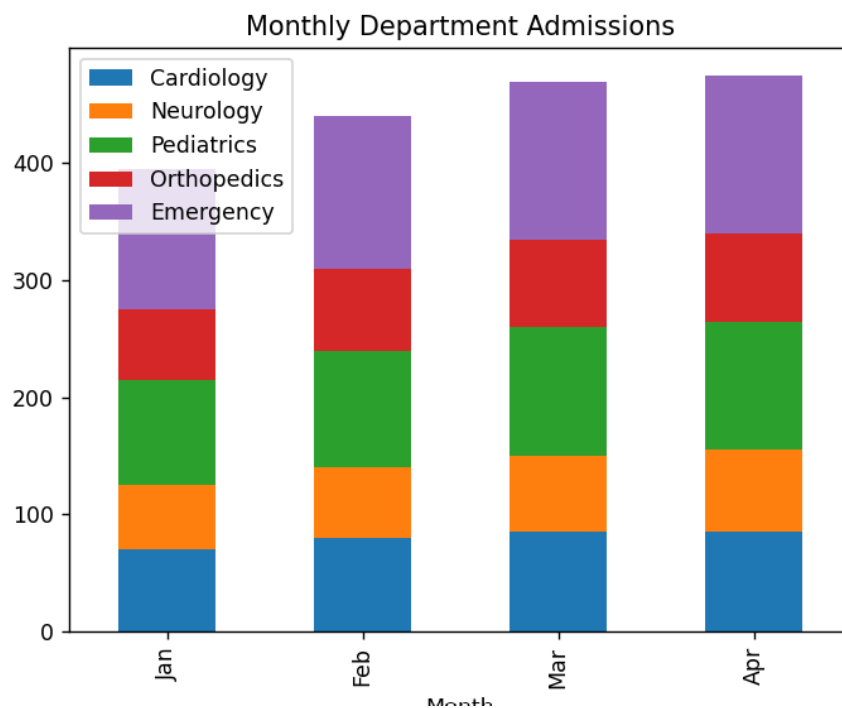
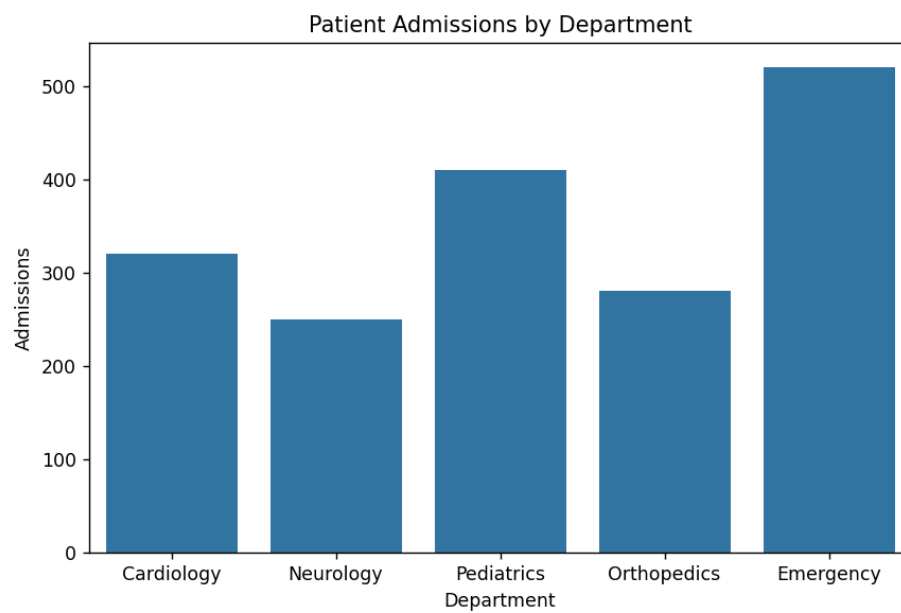
box_data = pd.DataFrame({
    'Department': ['Cardiology']*4 + ['Neurology']*4 + ['Pediatrics']*4 +
    ['Orthopedics']*4 + ['Emergency']*4,
    'Admissions':
    [70,80,85,85,55,60,65,70,90,100,110,110,60,70,75,75,120,130,135,135]
})

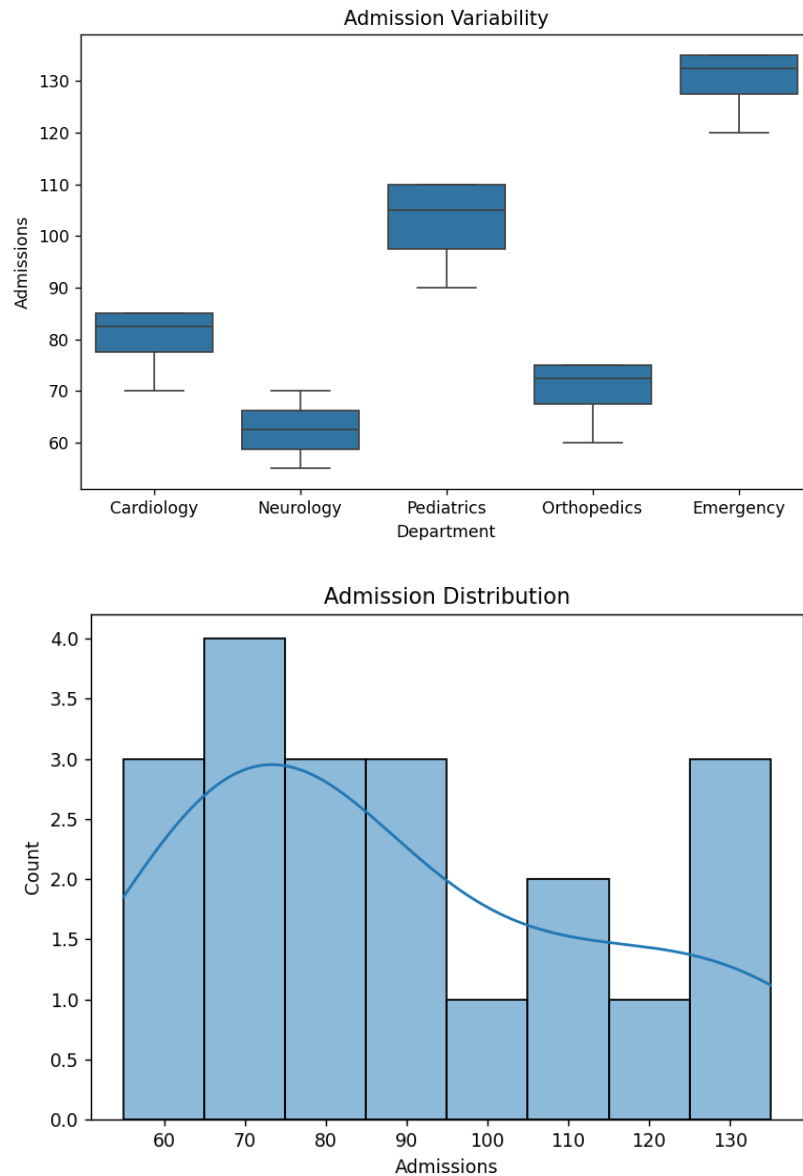
plt.figure(figsize=(8,5))
```

```
sns.boxplot(x='Department', y='Admissions', data=box_data)
plt.title('Admission Variability')
plt.show()
```

```
plt.figure(figsize=(7,5))
sns.histplot(box_data['Admissions'], bins=8, kde=True)
plt.title('Admission Distribution')
plt.show()
```

Output :





2. E-Commerce Customer Spending Behaviour

Code :

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
data = pd.DataFrame({
```

```
"PurchaseAmount": [120, 250, 180, 400, 320, 150, 500, 220, 280, 450, 350, 200, 600, 170, 390],
```

```
"OrderFrequency": [2, 5, 3, 8, 6, 2, 10, 4, 5, 9, 7, 3, 12, 2, 8],
```

```
"Category": ["Electronics", "Fashion", "Fashion", "Electronics", "Home", "Home", "Electr
```

```
onics","Fashion","Beauty","Electronics","Home","Beauty","Electronics","Fashion","Home"]
}))
```

```
plt.figure(figsize=(6,4))
sns.histplot(data["PurchaseAmount"], bins=8, kde=True)
plt.title("Histogram of Purchase Amount")
plt.show()
```

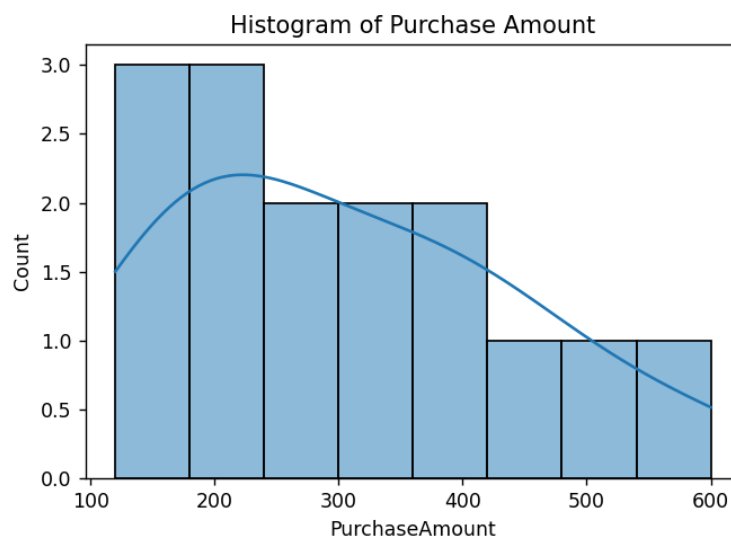
```
plt.figure(figsize=(6,4))
sns.violinplot(x="Category", y="PurchaseAmount", data=data)
plt.title("Violin Plot")
plt.show()
```

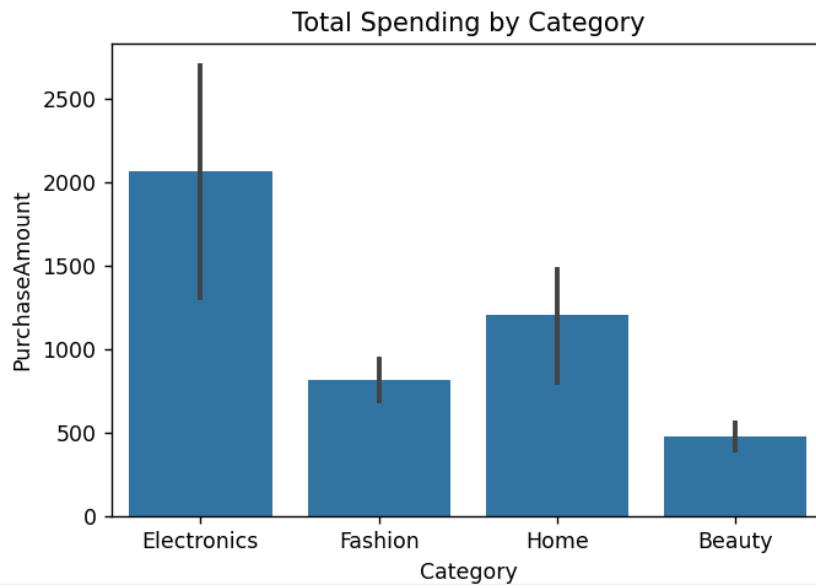
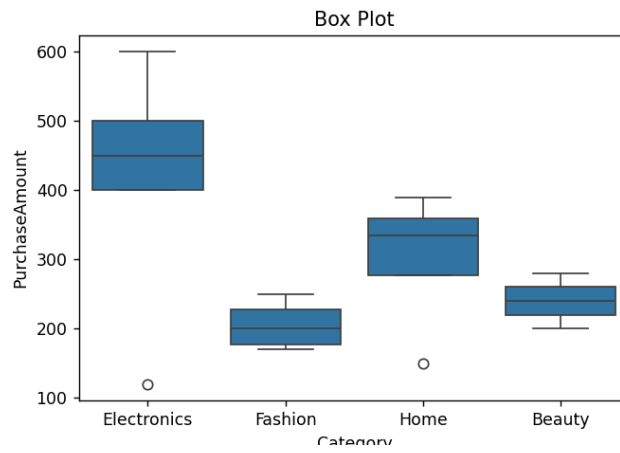
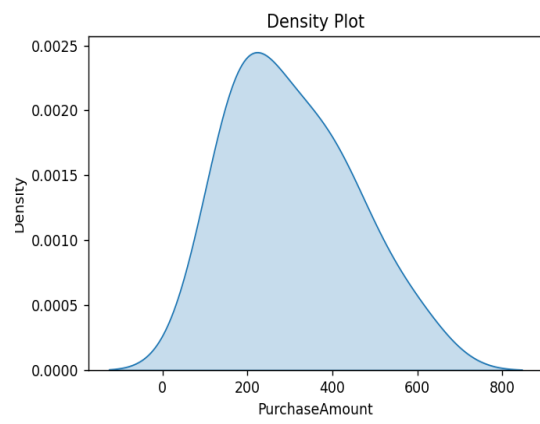
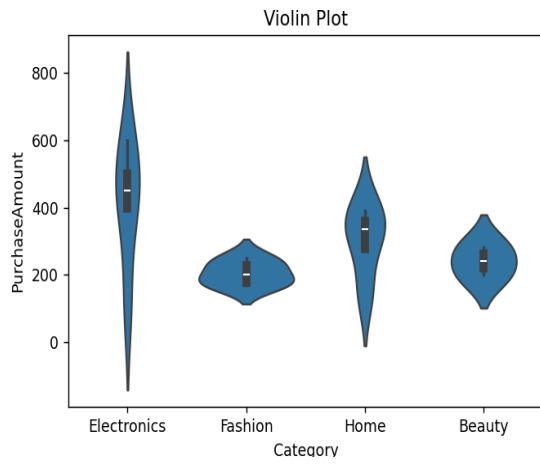
```
plt.figure(figsize=(6,4))
sns.boxplot(x="Category", y="PurchaseAmount", data=data)
plt.title("Box Plot")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.kdeplot(data["PurchaseAmount"], fill=True)
plt.title("Density Plot")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.barplot(x="Category", y="PurchaseAmount", data=data, estimator=sum)
plt.title("Total Spending by Category")
plt.show()
```

Output :





3. University Examination Performance Dashboard

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.DataFrame({

    "Department":["CSE","CSE","ECE","ECE","IT","IT","MECH","MECH","CIVIL","CIVIL"],
    "Class":["A","B","A","B","A","B","A","B","A","B"],
    "Attendance":[95,88,90,85,92,86,80,75,89,84],
    "Marks":[92,85,88,78,91,84,73,70,82,79]
})

sns.barplot(x="Department", y="Marks", data=df)
plt.title("Average Marks by Department")
plt.show()

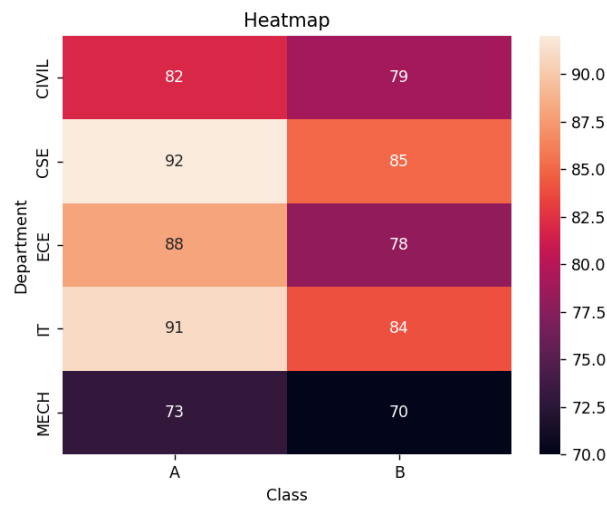
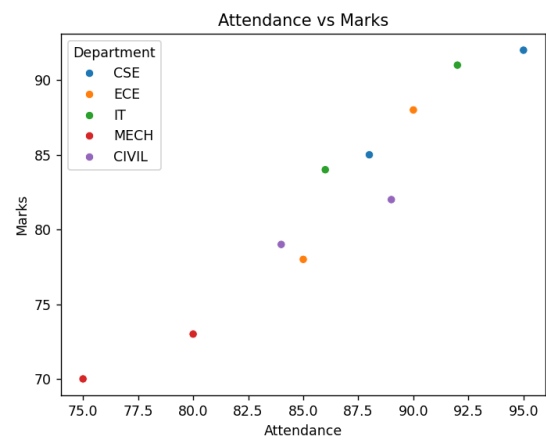
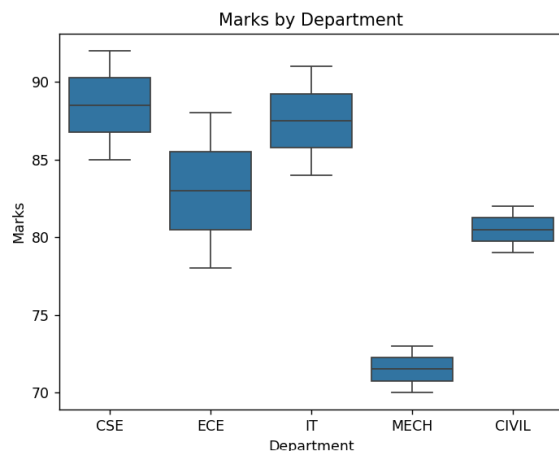
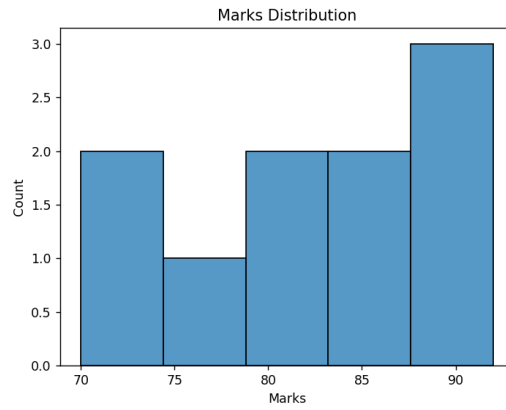
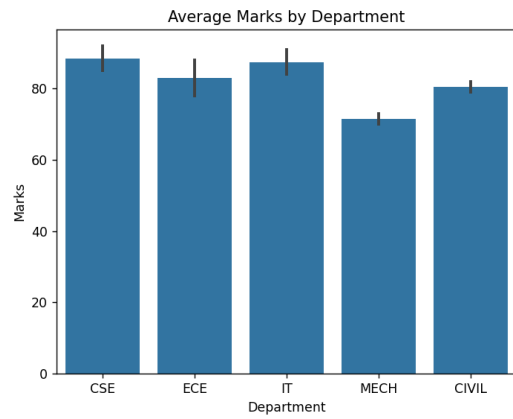
sns.histplot(df["Marks"], bins=5)
plt.title("Marks Distribution")
plt.show()

sns.boxplot(x="Department", y="Marks", data=df)
plt.title("Marks by Department")
plt.show()

sns.heatmap(df.pivot_table(values="Marks", index="Department", columns="Class"),
            annot=True)
plt.title("Heatmap")
plt.show()

sns.scatterplot(x="Attendance", y="Marks", hue="Department", data=df)
plt.title("Attendance vs Marks")
plt.show()
```

Output :



4. Climate and Rainfall Distribution Study

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.DataFrame({
    "City":["Chennai","Mumbai","Delhi","Bangalore","Hyderabad","Kolkata"],
    "Rainfall":[120,250,80,150,100,220],
    "Temperature":[34,31,39,28,33,30],
    "Humidity":[75,88,60,70,65,85]
})

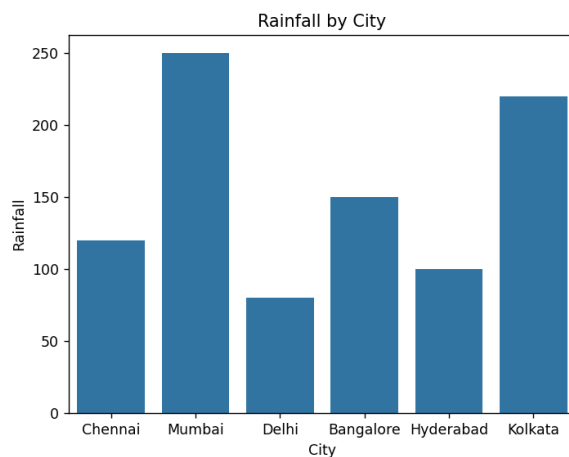
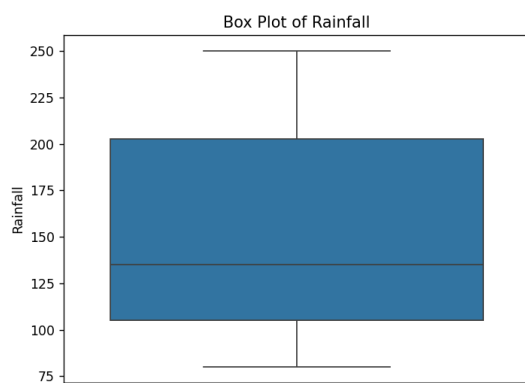
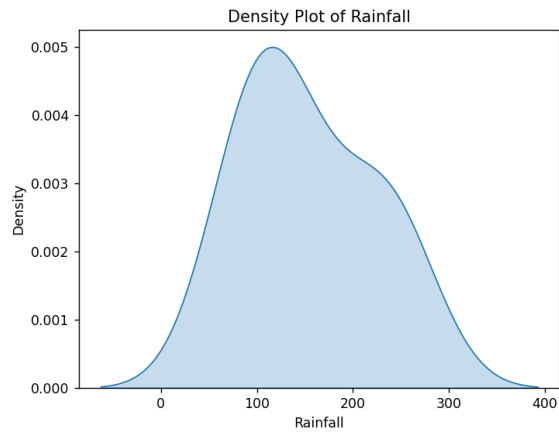
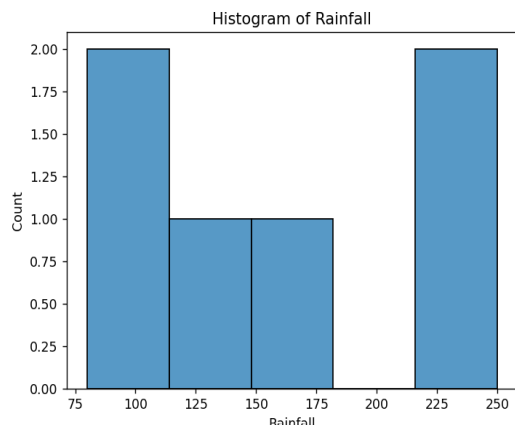
sns.histplot(df["Rainfall"], bins=5)
plt.title("Histogram of Rainfall")
plt.show()

sns.kdeplot(df["Rainfall"], fill=True)
plt.title("Density Plot of Rainfall")
plt.show()

sns.boxplot(y=df["Rainfall"])
plt.title("Box Plot of Rainfall")
plt.show()

sns.barplot(x="City", y="Rainfall", data=df)
plt.title("Rainfall by City")
plt.show()
```

Output :



5. Bank Loan Risk Assessment

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.DataFrame({
    "LoanAmount":[100000,150000,200000,250000,300000,350000],
    "Income":[30000,45000,50000,60000,70000,80000],
    "CreditScore":[620,680,720,750,790,810],
    "Repayment":["Poor","Average","Good","Good","Excellent","Excellent"]
})
```

```
sns.histplot(df["LoanAmount"], bins=5)
plt.title("Loan Amount Distribution")
plt.show()
```

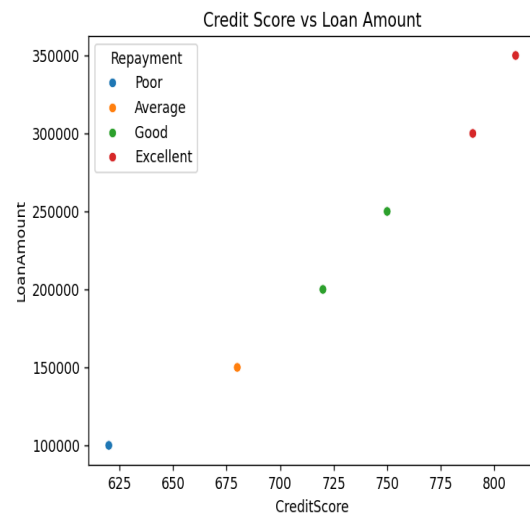
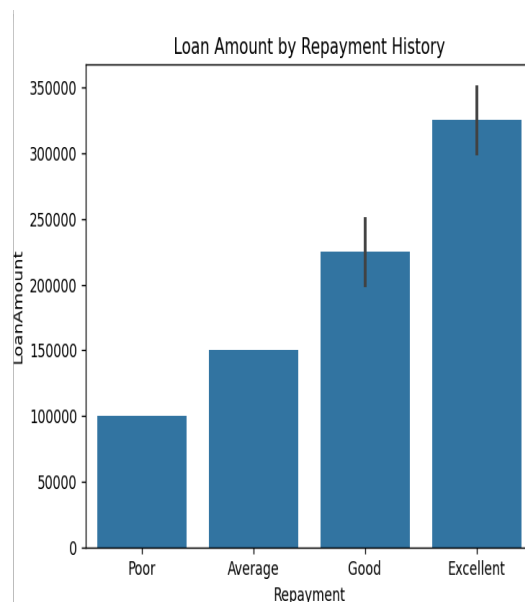
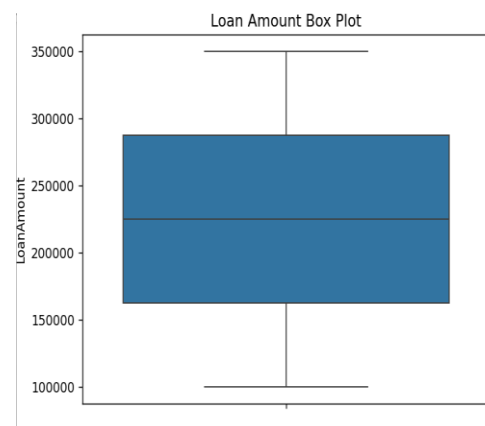
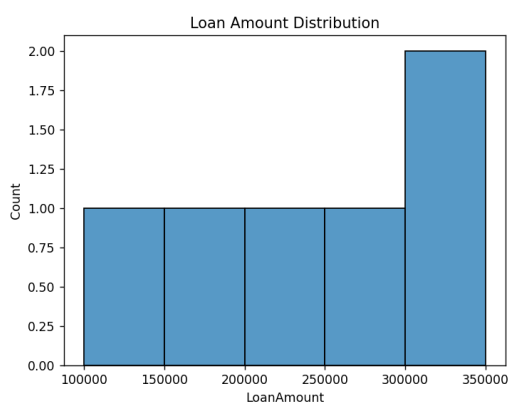
```
sns.boxplot(y=df["LoanAmount"])
```

```
plt.title("Loan Amount Box Plot")
plt.show()
```

```
sns.barplot(x="Repayment", y="LoanAmount", data=df)
plt.title("Loan Amount by Repayment History")
plt.show()
```

```
sns.scatterplot(x="CreditScore", y="LoanAmount", hue="Repayment", data=df)
plt.title("Credit Score vs Loan Amount")
plt.show()
```

Output :



Using R Programming

1. Hospital Patient Admission Analysis

Code :

```
library(ggplot2)

df <- data.frame(
  Department=c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
  Admissions=c(220,180,300,200,400),
  Male=c(120,90,160,110,220),
  Female=c(100,90,140,90,180)
)

ggplot(df,aes(x=Department,y=Admissions,fill=Department))+
  geom_col()+
  ggtitle("Bar Chart")

df_long <- data.frame(
  Department=rep(df$Department,2),
  Gender=rep(c("Male","Female"),each=5),
  Count=c(df$Male,df$Female)
)

ggplot(df_long,aes(x=Department,y=Count,fill=Gender))+
  geom_col()+
  ggtitle("Stacked Bar Chart")

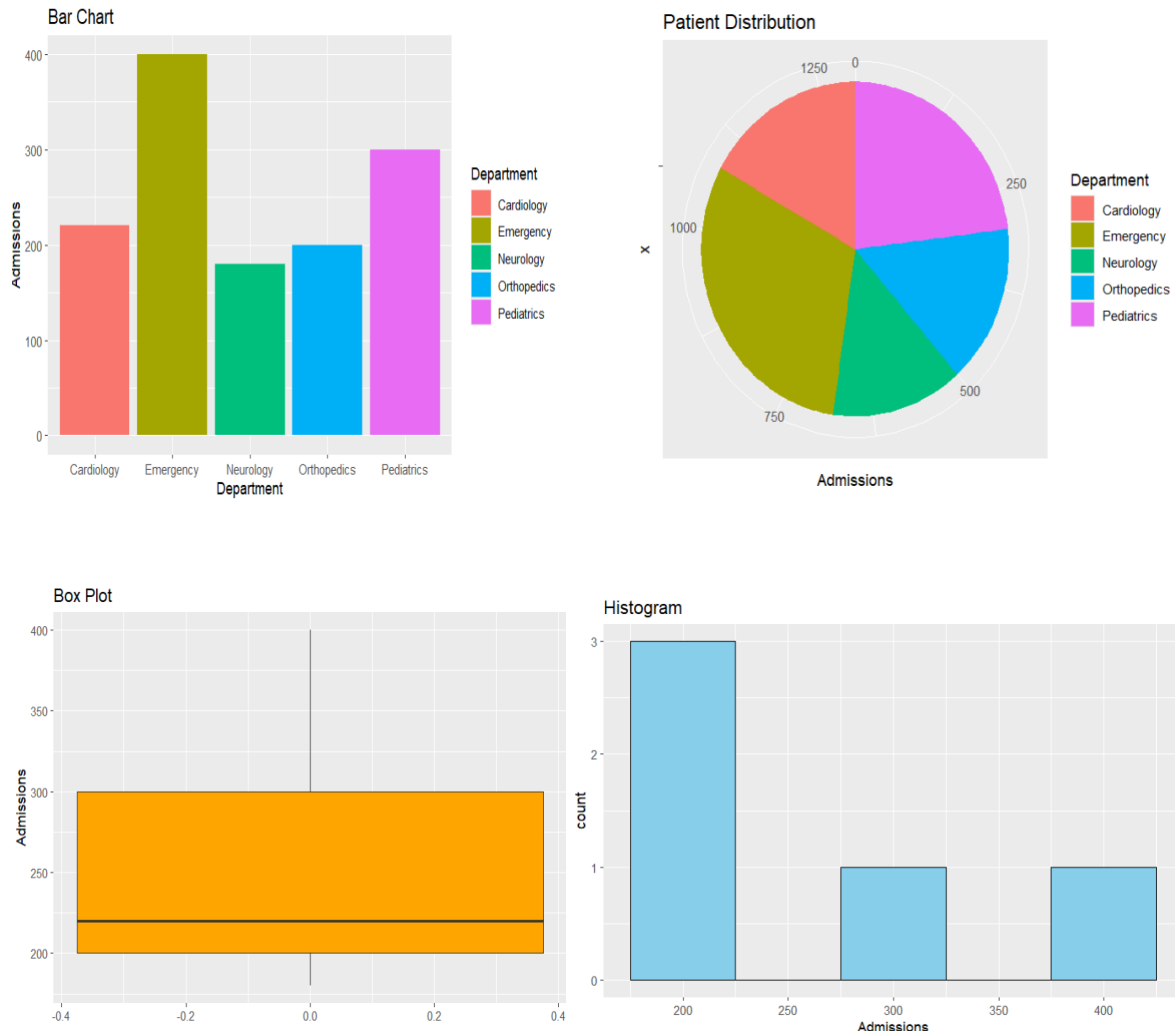
ggplot(df,aes(x="",y=Admissions,fill=Department))+
  geom_col(width=1)+
  coord_polar("y")+
  ggtitle("Patient Distribution")

ggplot(df,aes(y=Admissions))+
  geom_boxplot(fill="orange")+
  ggtitle("Box Plot")

ggplot(df,aes(x=Admissions))+
  geom_histogram(binwidth=50,fill="skyblue",color="black")+
```

```
ggtitle("Histogram")
```

Output :



2. E-Commerce Customer Spending Behaviour

Code :

```
library(ggplot2)
```

```
df <- data.frame(
  Category=c("Electronics","Fashion","Grocery","Books","Sports","Beauty"),
  PurchaseAmount=c(5000,3000,2000,1500,3500,2500),
  OrderFrequency=c(15,12,20,8,10,14)
```

)

```
ggplot(df,aes(x=PurchaseAmount))+  
  geom_histogram(binwidth=1000,fill="skyblue",color="black")+  
  ggtitle("Histogram")
```

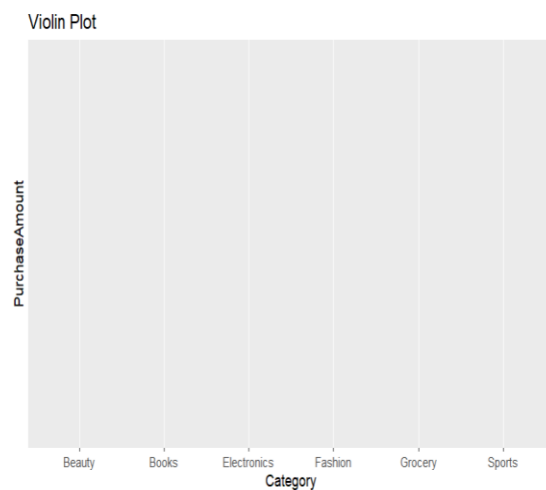
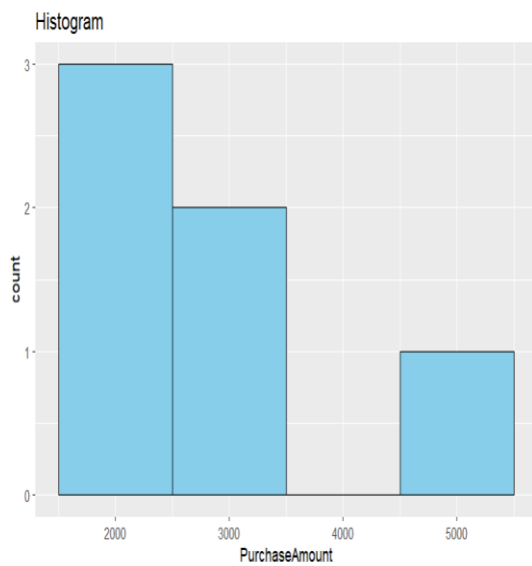
```
ggplot(df,aes(x=Category,y=PurchaseAmount,fill=Category))+  
  geom_violin()+  
  ggtitle("Violin Plot")
```

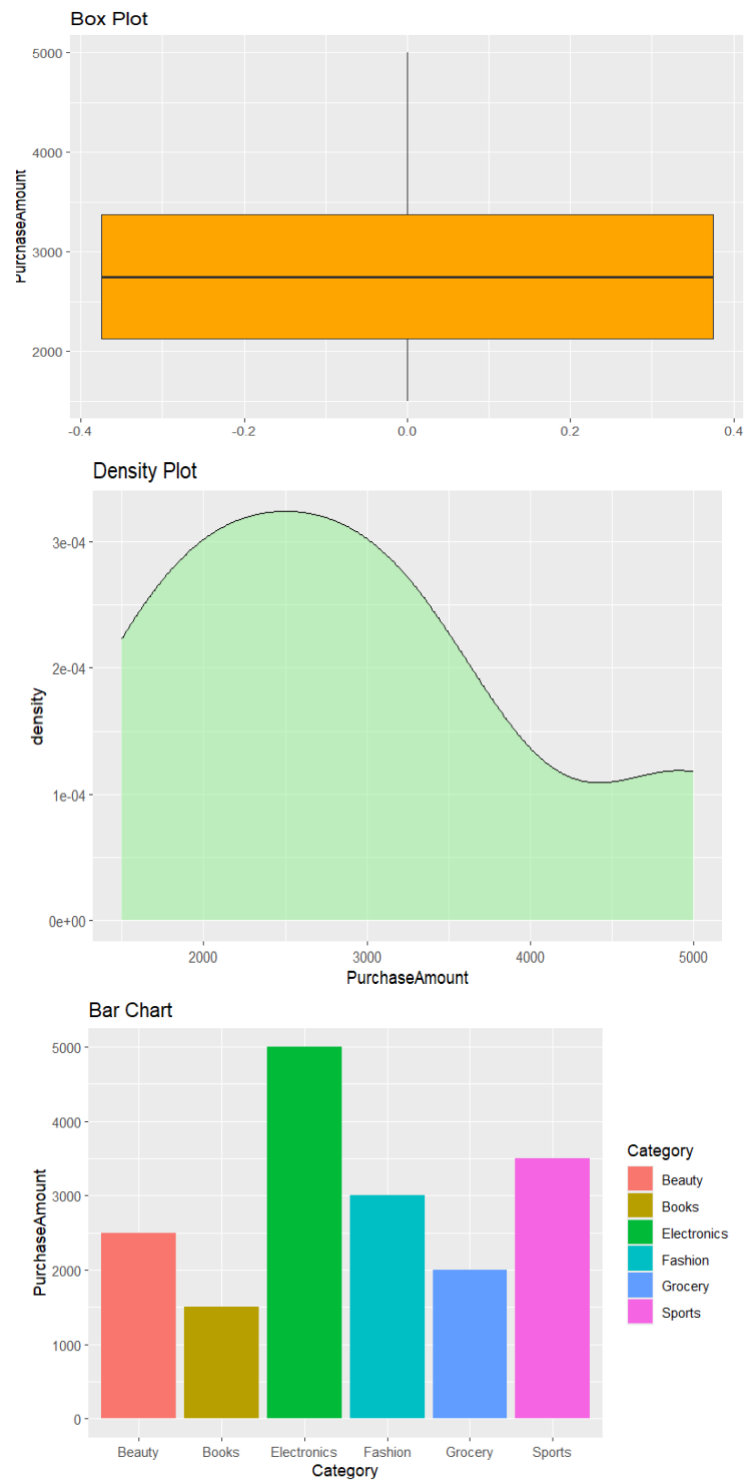
```
ggplot(df,aes(y=PurchaseAmount))+  
  geom_boxplot(fill="orange")+  
  ggtitle("Box Plot")
```

```
ggplot(df,aes(x=PurchaseAmount))+  
  geom_density(fill="lightgreen",alpha=0.5)+  
  ggtitle("Density Plot")
```

```
ggplot(df,aes(x=Category,y=PurchaseAmount,fill=Category))+  
  geom_col()+  
  ggtitle("Bar Chart")
```

Output :





3. University Examination Performance Dashboard

Code :

```
library(ggplot2)
```

```
df <- data.frame(
```

```

Department=c("CSE","ECE","MECH","CIVIL","IT"),
Marks=c(85,78,72,80,90),
Class=c("A","B","A","B","A")
)

```

```

ggplot(df,aes(Department,Marks,fill=Department))+
  geom_col()+
  ggtitle("Bar Chart")

```

```

ggplot(df,aes(Marks))+
  geom_histogram(binwidth=5,fill="skyblue",color="black")+
  ggtitle("Histogram")

```

```

ggplot(df,aes(y=Marks))+
  geom_boxplot(fill="orange")+
  ggtitle("Box Plot")

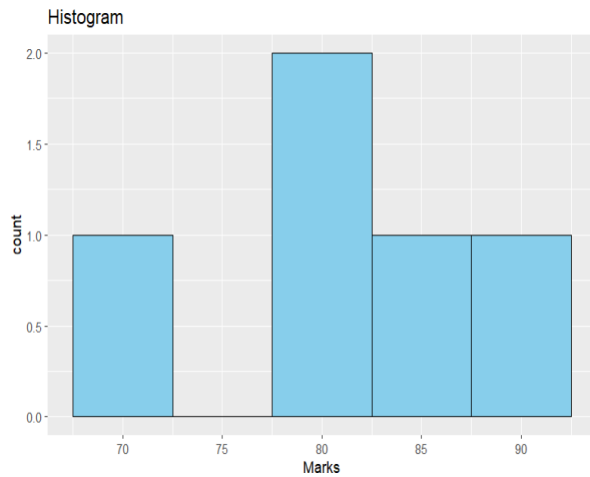
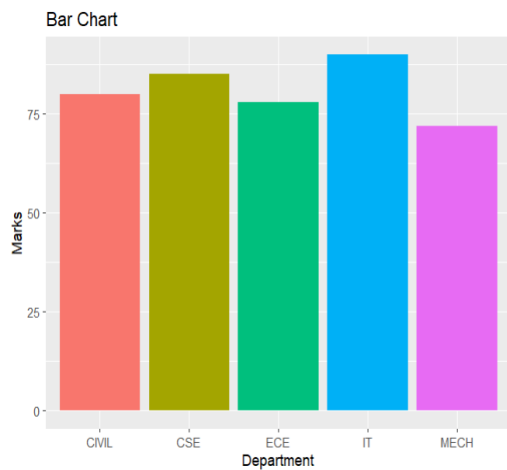
```

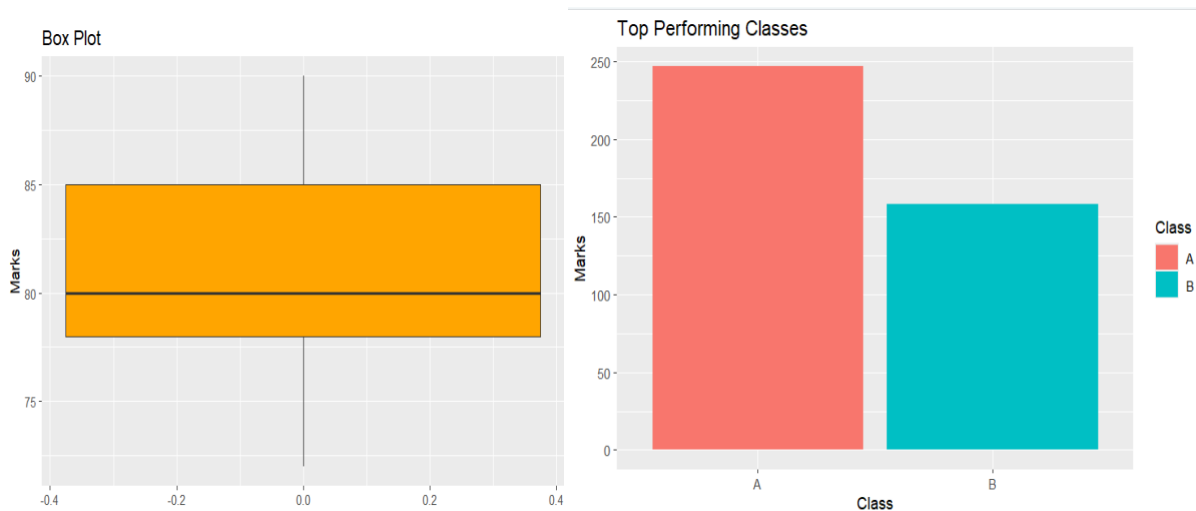
```

ggplot(df,aes(Class,Marks,fill=Class))+
  geom_col()+
  ggtitle("Top Performing Classes")

```

Output :





4. Climate and Rainfall Distribution Study

Code :

```
library(ggplot2)
```

```
df <- data.frame(
  City=c("Chennai","Mumbai","Delhi","Bangalore","Hyderabad","Kolkata"),
  Rainfall=c(120,250,80,150,100,220),
  Temperature=c(34,31,39,28,33,30),
  Humidity=c(75,88,60,70,65,85)
)
```

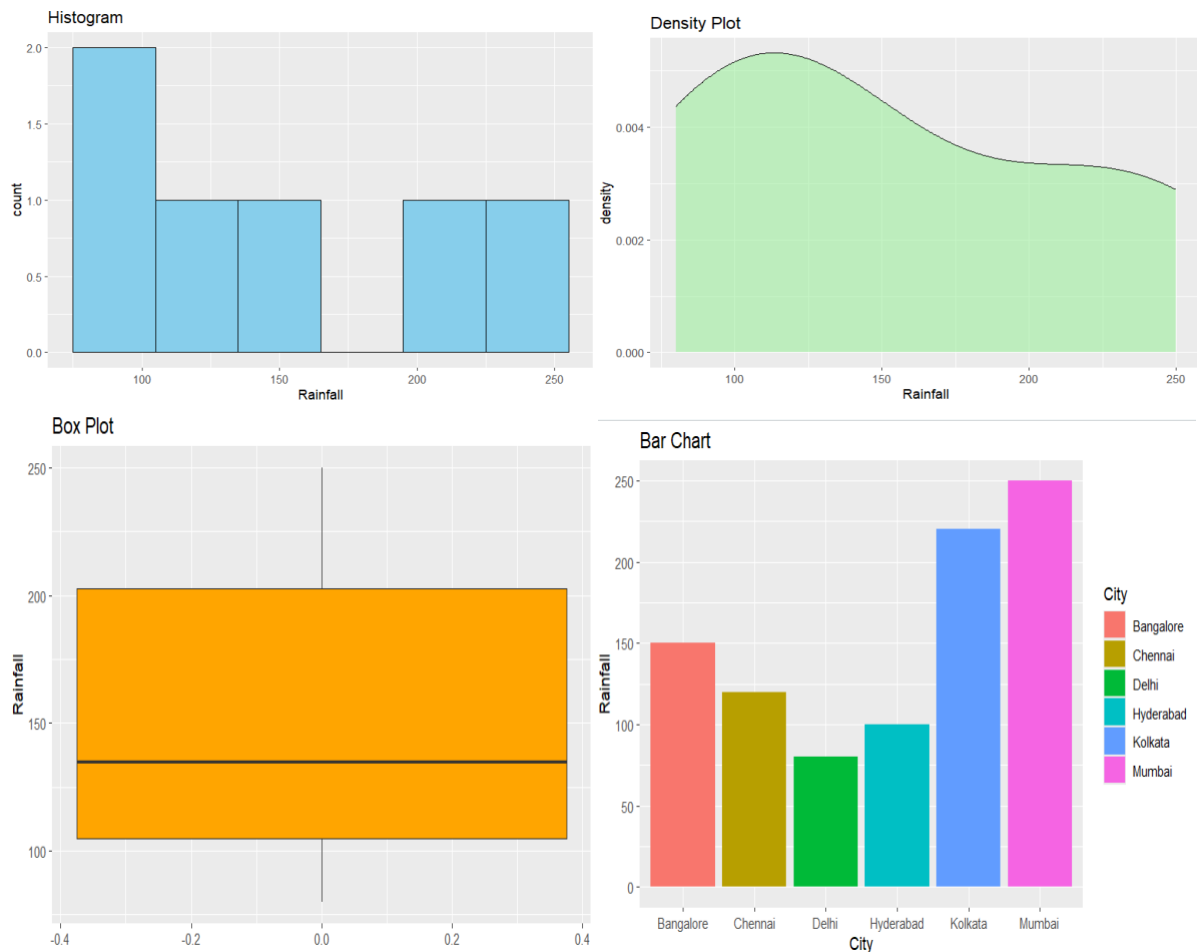
```
ggplot(df,aes(Rainfall))+
  geom_histogram(binwidth=30,fill="skyblue",color="black")+
  ggtitle("Histogram")
```

```
ggplot(df,aes(Rainfall))+
  geom_density(fill="lightgreen",alpha=0.5)+
  ggtitle("Density Plot")
```

```
ggplot(df,aes(y=Rainfall))+
  geom_boxplot(fill="orange")+
  ggtitle("Box Plot")
```

```
ggplot(df,aes(City,Rainfall,fill=City))+
  geom_col()+
  ggtitle("Bar Chart")
```


Output :



5. Bank Loan Risk Assessment

Code :

```
library(ggplot2)

df <- data.frame(
  LoanAmount=c(100000,150000,200000,250000,300000,350000),
  Income=c(30000,45000,50000,60000,70000,80000),
  CreditScore=c(620,680,720,750,790,810),
  Repayment=c("Poor","Average","Good","Good","Excellent","Excellent")
)

ggplot(df,aes(LoanAmount))+
  geom_histogram(binwidth=50000,fill="skyblue",color="black")+
  ggtitle("Histogram")

ggplot(df,aes(y=LoanAmount))+
```

```
geom_boxplot(fill="orange")+
ggtitle("Box Plot")
```

```
ggplot(df,aes(Repayment,LoanAmount,fill=Repayment))+
geom_col()+
ggtitle("Bar Chart")
```

```
ggplot(df,aes(Income,CreditScore,color=Repayment))+
geom_point(size=3)+
ggtitle("Customer Segments")
```

Output :

